Lesson 14: Raising the Stakes

Unreal Engine (optional advanced track)

VIDEO GAME CREATION

This week: later waves get genuinely harder - more enemies, moving faster - using the exact wave variable you've had since Lesson 11.

Open your own [YourName]Game project from Lesson 13 - don't start a new one.

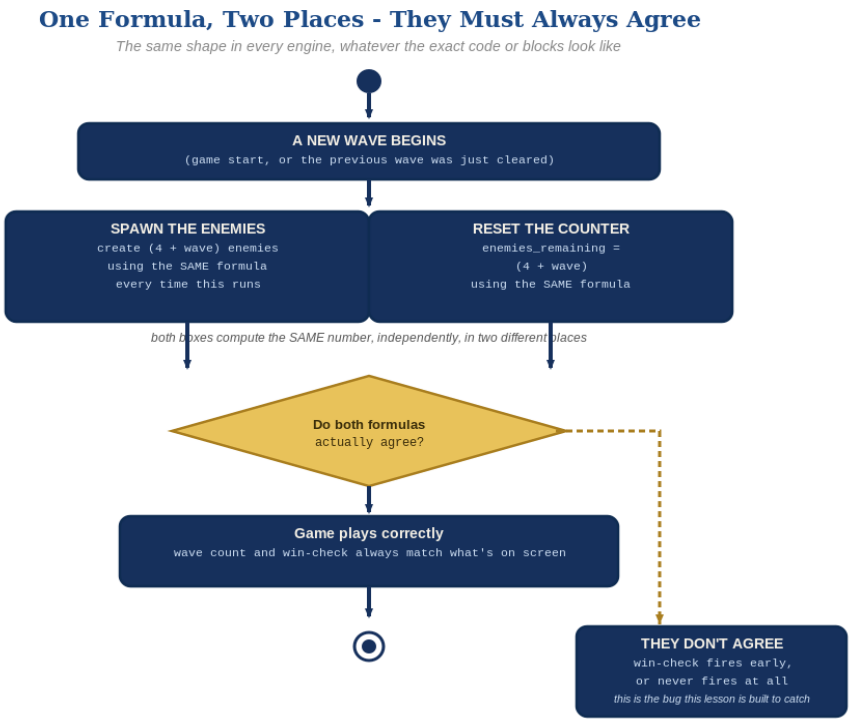
Tip: no new assets this week - difficulty scaling reuses your exact existing enemy sprites and spawn logic.

Setup

1. Open your [YourName]Game project from Lesson 13.
2. No new variables needed for Level 1 - Level 2 will use BP_Enemy's existing EnemySpeed variable (from Lesson 7).

Level 1 - Minimal (MVP)

Each wave spawns one more enemy than the last (5, 6, 7). Speed stays the same for now - that's Level 2.



Whether it's two places (most engines) or the loop and the reset living in genuinely different scripts (Godot, GameMaker), the rule is the same: one formula, referenced everywhere.

VIDEO GAME CREATION

Whether the spawn loop and the reset live in two places or two genuinely different scripts, the rule is the same: one formula, referenced everywhere.

3. In your SpawnWave Custom Event, change the fixed For Loop count of 5 into (4 + Wave) (code below) - wave 1 spawns 5, wave 2 spawns 6, wave 3 spawns 7.
4. Wherever EnemiesRemaining gets reset to 5 for a new wave, change that to 4 + Wave too - these two places must always agree.
5. Play in Editor and play through a full round - wave 2 should visibly spawn one more enemy than wave 1, and wave 3 one more than that.

```
Custom Event SpawnWave (revised) ->
  For Loop (0 to (4 + Wave) - 1) -> Spawn Actor from Class(BP_Enemy) ->
Bind Event to OnEnemyDestroyed / OnEnemyEscaped
  EnemiesRemaining = 4 + Wave
```

Alternate Path: C++ (code, optional advanced track)

The same wave-dependent formulas work as plain C++ expressions instead of Blueprint math nodes.

6. In SpawnWave(), replace the fixed loop bound of 5 with (4 + Wave), and set each newly spawned enemy's EnemySpeed to 50.f + (Wave - 1) * 15.f right after spawning it.

This mirrors the Blueprint path's For Loop and Set EnemySpeed nodes exactly - same formulas, just in code.

Level 2 - Complete (Alpha/Beta)

Enemy speed ALSO increases each wave, on top of the growing count. Restarting correctly returns everything to wave 1's original values.

- On each spawned BP_Enemy, right after Spawn Actor from Class, add Set EnemySpeed = 50.0 + (Wave - 1) * 15.0.
- Confirm BP_Enemy's own Event Tick movement (from Lesson 7) is still what actually reads EnemySpeed - no changes needed there.
- Restart mid-game and confirm both count and speed return correctly to wave 1's values.
- Play a full round and confirm wave 3 is noticeably busier and faster than wave 1, but still fair.

Level 3 - Refined - Choose ONE Stretch Feature

Pick exactly one of the five ideas below. These are intentionally open-ended - a good-faith attempt at the general idea is a complete Level 3 finish, not a fully polished feature.

- Destructible shields/barriers: a few simple blocks placed between the player and the enemies that disappear (or shrink) after absorbing a hit or two - reuses your existing collision-detection and hit-reaction patterns from Lessons 4 and 13, just on a new kind of object.
- A power-up: an item that occasionally drops or spawns, which the player collects to temporarily change how shooting works (faster fire rate, multiple bullets at once, or a brief shield) - reuses your existing timer/cooldown pattern from Lesson 3.
- A bonus enemy: a rare, faster-moving, or oddly-patterned enemy worth extra points, spawned occasionally alongside a normal wave - reuses this week's own speed/count scaling ideas, just for one special case instead of the whole wave.
- Formation-marching movement: instead of each enemy independently drifting down (this course's approach since Lesson 4), have the whole wave move together as a group and occasionally step sideways - a genuinely different movement pattern, inspired by classic arcade shooters.
- A title screen and/or game-over screen: a proper opening screen (press a key to start) and/or a dedicated game-over screen with a final score, instead of just text overlaid on the gameplay - reuses your existing game_active flag and UI tools from Lessons 9-10. Unreal's Widget Blueprint system (already used for your HUD since Lesson 9) is a natural fit for a title/game-over screen.

If Something's Wrong

- Every wave still spawns exactly 5: confirm the For Loop actually uses (4 + Wave) - 1 as its end value, not a leftover literal 4.
- The win message fires early or never: confirm EnemiesRemaining is reset using the SAME formula everywhere it's set.
- Speed doesn't change between waves: confirm Set EnemySpeed is wired right after each Spawn Actor from Class node inside SpawnWave, not left at BP_Enemy's default forever.

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